Title: Complementing Collaborative Learning with Generative Artificial Intelligence

Abstract: I describe an approach for integrating artificial intelligence (AI) into economics courses in a way that promotes critical thinking and fosters AI literacy skills. Many instructors are apprehensive about AI because it raises questions about integrity, ethics, and equity. I modify the send-a-problem collaborative learning technique in which students first develop foundational knowledge individually, then probe that knowledge by joining teams and using AI to create complex problems, then finally solving the problems created by their peers. Student reflections suggest the activities are engaging and beneficial. Examples of prompts and problems created by students are included.

1. Introduction

Artificial intelligence[1] (AI) is rapidly infiltrating our lives and changing the way we do many things. It is abundantly clear that students are using AI for assignments and studying, in large numbers, with the intensity of use almost sure to grow (Staveley-O'Carroll et al., 2026). The teaching profession is not immune to this technological change, and it is imperative upon us, as instructors, to educate ourselves about the technology so that we may integrate it effectively into our course planning. For some instructors, this may mean intentionally designing a class or assignment that does *not* use AI. For others, this may mean incorporating multiple touch points where students use some form of AI for deliberately chosen tasks. Others may take the approach of permitting any use outside the classroom, while moving assessments to in-person, pencil-and-paper endeavors. Regardless of one's chosen approach, AI has evolved such that it must be a consideration when planning a course.

One pressing concern is that students may use AI in ways that are counterproductive or harmful to learning. For example, Staveley-O'Carroll et al. (2026) collect survey data from over 2,300 economics students across 24 institutions and find that students report using AI to solve math problems, answer conceptual questions, and help complete exams. They also find that one-third of students think it is ethical to use AI to complete assignments regardless of the instructor's policy. This points to the need for instructors to design their courses with incentives that encourage students to embrace the productive struggle of the learning process, while avoiding the pitfalls of using AI as a shortcut. One potential avenue for capturing some of the benefits of AI while minimizing its potential costs is with cooperative learning (Marburger, 2005). In particular, the

[1] When referring to AI, I generally have in mind any type of generative artificial intelligence and, specifically, I am thinking about any type of chat-based application that students can use to receive direct responses to their queries.

focus of this article is the application of a collaborative learning technique, known as send-a-problem, in conjunction with AI to elicit critical thinking about economics concepts and AI itself.

The extent to which an instructor chooses to use AI to complement the learning process should depend in part on the existing knowledge base of the student. For example, students in an introductory economics course likely have little to no foundational understanding of the core concepts of the field, especially at the beginning of the course. As such, using AI may be more hindrance than helpful to them, if used incorrectly. For instructors, this means that assigning traditional homework and papers to be completed outside of class may not provide a true indication of a student's level of mastery. It is possible, perhaps even likely, that students will submit AI-generated answers to such problems, as those answers are now much easier to acquire and remarkably accurate. While some instructors may claim that the latest AI cannot get an A on their exam/assignment, we must remember that all students have different preferences, and many may optimize on simply doing what is needed to pass the course. AI can easily provide passing answers in most courses (Geerling et al., 2023), especially if the course is structured poorly, and AI's performance is likely to continue improving rapidly.

There are ways in which AI can be a helpful tool. For example, students may be able to use AI outside of class to test themselves, create study guides, and generate practice problems (Staveley-O'Carroll et al., 2026). However, students need to have a strong foundational understanding of the concepts to ensure they are using the AI tools correctly (Klein and Klein, 2025). One way to help students build that foundation is by using evidence-based learning strategies such as send-a-problem (McGoldrick et al., 2010) in conjunction with AI. This can serve the dual purpose of improving AI literacy while simultaneously enhancing economic literacy.

Furthermore, this helps model proper AI use for students and provides some practical application of what they are learning.

Send-a-problem is a collaborative learning technique that has been around for decades and was most recently popularized by Barkley, Major, and Cross (2014).[2] With this technique, an instructor first places students into teams or teams. Then, each team is given an envelope or folder with a different, complex problem attached to the outside of the envelope/folder. Students work together to solve all or part of the problem and place their solution inside the envelope/folder. After a pre-determined amount of time, the teams "send" the problem to another team, who then try to solve the same problem and likewise puts their solution into the envelope. After a few rounds, depending on how many teams you have and how long your class periods are, each team opens an envelope and assesses the attempts inside to see which answers are the closest to correct. At this point, most of the teams will have attempted to solve many, if not all, the problems except for the one they are currently grading.

The send-a-problem exercise offers many potential benefits. First, it provides students with practice applying course concepts which can lead to deeper, permanent learning, consistent with taxonomies from Bloom (Bloom, 1956) and Fink (Fink, 2003). Second, it provides an opportunity for peer learning and peer teaching, which is another evidence-based best practice (Mazur, 1997). Third, it encourages metacognition because students must think about the problem and the answers offered by their peers. This requires them to interrogate the way they think about the problem relative to someone else which, again, can lead to more permanent learning (Young and Fry, 2008). Fourth, a cooperative learning exercise is, by its nature, a social endeavor and that helps students

[2] I use the terms collaborative learning and cooperative learning somewhat interchangeably, wherein I am simply referring to students working together to complete a common task. However, some have made the distinction that cooperative learning is a more structured form of collaborative learning (pp. 5-7, Barkley, Major, and Cross, 2014).

learn to work in a team environment, fostering a sense of trust and belonging in the classroom (Beattie and Ersoy, 2025).

I modified this assignment to hopefully create an additional benefit. Rather than giving students a problem to solve, I provided the students with guidance for creating an AI prompt so that they can generate a problem themselves with AI assistance. AI is a tool and that means it is only as useful as the person wielding it. Further, creating opportunities for building AI literacy is important and increasingly valued by employers, as use among employees continues to grow (Crane et al., 2025; Gallup, 2026). In using AI to create the problem themselves, students must employ critical thinking skills to ensure the problem is solvable and achieves the learning goals set forth in the instructions.

2. The Context

My institution caps individual class sizes at 33 students and I usually have 27-33 students per section. This allows me to create seven or eight teams of students in each section. However, this could also take place in a large lecture because the primary feature of the activity is students working together in a small team on a complex problem. The send-a-problem technique is versatile so that you can "send" the problem as many or as few times as needed to achieve your learning objectives and within your classroom context. My classes meet for 100 minutes twice per week and I usually allocate at least 50 minutes for this activity, but the assignment is flexible enough to fit in a shorter or longer time frame, depending on how complex the problems are that you want students to solve and how many times teams "send" the problem. Finally, the technique works for introductory courses as well as upper-level electives. I used this technique in my Principles of Economics and my Intermediate Macroeconomic Theory courses. While versatile, the send-a-problem technique tends to work best when the problems are relatively complex. For example, topics involving simple calculation or rote memorization might result in a less successful exercise. I try to use it when pulling many concepts together, such as multiple shifts in supply and demand, sometimes in conjunction with curves of different elasticities. It also works well when revisiting previously covered material and applying it in a new context, such as supply and demand in the context of externalities. Table 1 provides examples of topics which may be conducive to the send-a-problem technique.

Table 1: Potential Topics to Use Send-a-Problem

Principles of Economics	Intermediate Macroeconomics
Calculating Opportunity Costs and Comparative Advantage	Measuring the Macroeconomy, including basic measures of GDP, Inflation, and Unemployment
Supply and Demand with shifting curves, calculation of equilibrium, elasticity	Solow Model
Market Structure and elasticity	IS-MP-PC Model
Externalities and Government Intervention	AD-AS Model
AD-AS Model	

An important step for any active learning activity is the development of learning goals. This helps ensure the assignment is focused and adds transparency for the students, so they understand they are not doing "busy work." The overarching goal of the activity is to provide a framework within which students can build mastery of the material. The general goals adopted for this assignment apply elements from Bloom (1956) and Fink (2003) and are intended to be concise so that students find them valuable and easy to understand. Those goals are as follows:

- Identify the key components of [concept or idea]
- Apply economic reasoning to create, revise and improve practice problems

- Analyze and critique AI-generated content for accuracy and relevance
- Create and solve original problems related to [concept or idea]

The assignment is designed to help students move from consumers to producers of economics practice problems. Doing this means students need a certain level of mastery so the problem they create is consistent with the theories of economics. Becoming a producer of such problems pushes the student closer to mastery than simply completing problems created by the instructor.

3. The Setup

I used this technique in two different courses: Principles of Economics and Intermediate Macroeconomics. However, it can be used in any economics course. General guidance on creating examples is provided in this section, with specific examples from each course and examples of student work provided in the Appendix. The first exercise in my Principles of Economics course was implemented early in the semester right after students learned the basics of demand, supply, and equilibrium. This ensured students had a foundation from which to build expertise. Students were given advice on creating a prompt for the AI tool of their choice.[3] Next, teams were given about fifteen minutes to chat with the AI until they were confident the example they created was correct and met the goals set forth in the instructions. Each team wrote their problem on a sheet of paper by hand, taped it to the front of a folder provided by the instructor, and then "sent" it to another team. The receiving teams were then given about fifteen minutes to solve the problem by hand and place their solutions inside the folder.

An important point here is that after AI was used to aid in the creation of the initial problem, computers and phones had to be put away, and problems solved by hand. This ensures students are

[3] I encouraged the use of Microsoft Copilot, for which students have access through the university, and ChatGPT because I am familiar with it and have a university license. The entire assignment, including the instructions given to students, is available in the Appendix.

still doing the heavy lifting of learning the material. It would be hard to ensure students were not also using AI to solve the problems if their computers or phones were readily available during this stage. It is also important to ensure students cannot use technology as a learning crutch or to opt out of socializing with their classmates on their shared learning effort. The "sending" stage can be repeated as many times as desired based on the length of a class, the number of teams, and the level of desired detail in the answers. Problems should be sent a minimum of two times, with a maximum equal to the number of teams minus one. The teams should be cautioned not to look at the answers of the previous teams.

On the last "send" the teams should open their folder or envelope and assess the accuracy of the solutions provided. At this stage, there should be at least two different solutions provided and there are different ways to assess the solutions. One way is to simply rank the solutions in order of the most accurate. Another way is to have students use a rubric to assess the work of their classmates. This could be helpful if the rubric is similar, for example, to how you grade exam questions. Students receive more information about how you assess their exams and have a better idea of the depth their answers should include. Depending on the classroom context, each team could also write a solution on the board or otherwise share the best answer with the class.

Alternatively, teams can add their solutions to a shared online document so that the instructor can assess each solution and provide feedback if necessary. This serves the dual purpose of creating an answer key for each of the problems created. Students now have several examples they can use as study material for exams or that instructors can use for review sessions or office hours problems. This may be especially useful for upper-level courses, where it can be more challenging to create fresh problems. Another possibility is that students could create an answer key, then have AI create an answer key, and compare their solutions for discrepancies.

Finally, students were asked to reflect on the experience of using AI to help create problems with a short, open-ended question asking if they found the exercise useful. Students provided these reflections after the class ended and their responses were graded based on completion only. This is an important part of the process as it reinforces for students that you care about their learning and want to create activities that they find helpful. Creating this buy-in early is a good way to foster an inclusive classroom environment and helps make the material more relevant to students (Al-Bahrani 2022).

4. Outcomes

As students were crafting their initial problems, I spoke with each team to monitor their progress, pose Socratic questions about their choices, and offer encouragement. Most teams were able to craft a relatively effective question that met the learning goals, but there were some teams who did not entirely understand what they were expected to do. This was due to a combination of a lack of preparation on their part and assignment instructions that were too vague for students at that level of understanding. This problem diminishes with repetition because I was better and providing instructions and students became more familiar with the format.

I did not empirically test whether this exercise improved learning outcomes, but I did ask students whether they enjoyed the activity and found it helpful. Most students (69%) reported enjoying the activity, noting that they liked working with their peers, designing the problem was helpful for learning, and practicing problems other teams created was useful. Examples of student reflections include:

> "Enjoyed doing this activity because I was able to better understand the material...brought me and my team closer together."

> "Really liked doing the practice problems with AI in class...helped me understand the material better."

> "I really like it because now I realize I can use AI to help create practice problems for my studying."

> "Really enjoyed the team assignment because it taught me how to use ChatGPT to help me study and solve real practice problems...I think we should do that same activity again!"

This demonstrates that incorporating AI helped some students increase confidence in their thinking and they perceived a better understanding of the material. Another benefit, admittedly unanticipated, is that it demonstrated to some students for the first time the potential of AI as a study aid. If used with proper scaffolding and feedback loops, AI can be a tool to improve learning. Another somewhat unexpected benefit came from the students who said it helped bring their team closer together.[4] While this is something we usually hope teamwork will do, sharing the common experience of using an AI tool to achieve a goal provides an opportunity for improving belonging and self-efficacy. However, it is important to note that feelings of learning do not necessarily translate to real learning gains (Deslauriers, 2019), so much more work is needed on the efficacy of AI as a learning tool.

There were a few students who expressed indifference, hesitancy, or resistance towards using AI. Of these resistant and hesitant students, some expressed concern about using AI, explaining that they typically do not trust or simply try to avoid it in general. For example:

> "I try not to use Ai when I can so I probably won't be doing it in my free time but as a class activity I found it to be helpful."

> "I enjoyed the practice and thought it was an interesting idea, however I try to avoid using AI so I didn't love it."

[4] My courses usually feature team-based learning, so that students are placed into teams early in the semester and keep the same team throughout. That is not necessary for this activity, though, because ad hoc teams can work just as well. Indeed, some students may even enjoy getting to meet a larger proportion of their classmates by switching teams up occasionally.

A few students said that they would rather the instructor make the problems, which seemed to be rooted in a combination of trust in the instructor and lack of confidence on the part of the student. Students may also simply perceive a high opportunity cost of using an unfamiliar tool to create practice problems. For the latter, this reinforces the need to be transparent with students about why you are using this method and how some amount of difficulty is desirable to form more permanent learning connections (Fries et al., 2021).

> "I liked the team part of it, but would rather you just create the problems and us solve it as a team. The problems weren't that great and had a couple errors in them."
>
> "I would have preferred if the professor had made the problems...Some teams did not make great problems, and much room was left for interpretation."
>
> "Working in teams was super helpful and inciteful. I felt the AI generated practice problems were not really worth the trouble, but I definitely would like to do more team work."

Students were asked a brief, open-ended question about whether they liked the activity and why. Notably, none of the students raised specific ethical concerns. However, this is an area which deserves further study. It may be the case that students did not want to raise ethical concerns because they were afraid of getting a bad grade, pressure from their peers, or because they genuinely had no concerns. It may also be the case that students are not aware of ethical issues involved in the usage of AI.

5. Discussion

Intentional integration of AI into courses does not necessarily mean that instructors and/or students use AI to complete tasks. Intentional integration may begin with the revision of course assignments, so that assessments are completed in person rather than outside of class. In this case, AI may not be used at all, but AI is still integrated into the course design and structure. While one could certainly find ways to incorporate the use of AI, as demonstrated in this paper, it may be premature to require the use of AI in a course due to several limitations.

First, although it is quite easy to find a freely available AI, much more powerful models are behind paywalls. Therefore, requiring students to use AI for a grade could be measuring students' unequal access to resources rather than ability, skill, or knowledge. This is true even if your institution provides access for a particular AI because, again, well-resourced students may still be able to access much more sophisticated models. As models improve, this limitation may diminish but it is always worth thinking about accessibility.

Second, using AI is rife with ethical issues, some of which students may not be aware. For example, there is evidence that some models have been trained by exploiting low-wage workers in developing countries (Du & Okolo 2025). Further, many models may have been trained on copyrighted materials (Longpre et al. 2024). Though using copyrighted materials is sometimes allowed for educational purposes, such circumstances usually require that you cite the original source you are using which is not always possible depending on how you interact with AI. An additional ethical concern is the environmental cost of the these systems. The environmental costs of AI are always changing and, while subjective in some respects, require another layer of consciousness and attention when using AI with students.

Finally, one must be very careful in how AI is used, with special attention given to students in the early stages of the learning process. Students learning new concepts need to struggle through so-called desirable difficulties to ensure they develop the foundation and critical thinking skills required to use AI effectively in the future. There is growing evidence that cognitive offloading can happen in a way that potentially undermines the learning process (Lodge and Loble 2026).

Despite the limitations and concerns surrounding AI, we must continue to innovate and embrace technological advances. Brynjolfsson, Chandar, and Chen (2025) present evidence that the proliferation of AI is already having a negative impact on job outcomes for recent graduates and those with jobs most exposed to AI, while other studies suggest this change may have begun even prior to the release of ChatGPT (Frank et al. 2026). Properly preparing students to enter this new labor force means teaching students how to learn because it is a skill that will be required of them more than perhaps any other.

AI offers incredible potential for those seeking to learn, but only if used properly and with a sufficient baseline of knowledge. One potentially fruitful way to combine both is integrating exercises into our courses where students practice using AI to teach themselves and then intentionally reflect on whether AI achieved what the student asked it to do. The send-a-problem exercise described here scaffolds that process by building the economics foundation first, then using AI to test understanding of those concepts. Once the problem is created, there is another feedback loop whereby students attempt to solve the problems created by other teams without the aid of AI. This can help identify ambiguity in the problems created, which leads to further discussion of the underlying concepts and how they are properly applied.

There are a few other potential benefits from incorporating such an activity. First, as more instructors seem to be moving homework into the classroom, this provides a rigorous exercise that requires relatively minimal preparation. Second, it provides an opportunity for students to improve their discernment skills, which is increasingly important in this new AI era. Finally, it can provide a foundation for future office hours visits by students. The classroom environment during these exercises show the instructor can be helpful and welcoming, and the content of the problems offer a talking point for students otherwise unsure of how to approach office hours. Future researchers should consider designing randomized controlled trials to test this or other exercises, while giving careful attention to the value added, or perhaps not added, of the AI component of the assignment. While grades and learning are outcomes of interest, measures of AI literacy and academic mindset are also worth assessing.

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Appendix

<u>*AI-augmented Assignment Example for Principles of Economics:*</u>

By the end of this activity, you will be able to:

- Identify the key components of a supply and demand problem involving market equilibrium
- Apply economic reasoning to revise or improve practice problems
- Analyze and critique AI-generated content for accuracy and relevance
- Create and solve original problems related to equilibrium

Instructions: Use AI (e.g., ChatGPT, Microsoft Copilot, etc.) to generate a supply and demand example that is relevant to something in the life of one of your team members. For example, maybe someone is interested in playing sports, attending concerts, or perhaps you can use something as simple as a favorite restaurant or type of food. Your task is to create one demand and one supply equation for your chosen market that can be used to solve for market equilibrium. Try to make the equilibrium numbers as realistic as possible. Next, provide a scenario in which both the supply and demand curves shift, explain the direction and reasoning for the shifts, and explain what happens to equilibrium price and quantity as a result. The shift does not have to involve new calculations. Rather, focus on the direction of the shifts. Finally, discuss how elasticity of each curve might influence the new equilibrium.

Result: You should have a problem with four parts:

(a) Create a scenario for which students must draw a supply and demand diagram.
(b) Solve for equilibrium price and quantity, then show this point on the diagram.
(c) Create a modification of the previous scenario in which demand and supply are impacted. Identify which way the curves shift and why they shift.
(d) Explain what happens to equilibrium price and quantity because of the shifts. You don't need to solve a new equation but instead describe whether the new price and quantity are higher/lower than the original equilibrium and why.
(e) Explain how elasticity matters in this context. In other words, how would the equilibrium be different if the elasticity of the curves were different?

Examples of Principles of Economics problems created by students

1. Part A: In the market for Elon sweatshirts, find the market equilibrium when $Q^D=2000-66.7P$ and $Q^S=66.7P$.

   Part B: In 2025, cotton becomes more accessible and Elon enrollment increases. Draw what would happen to the supply and demand curves. Draw the new equilibrium on the graph.

   Part C: What happens to the equilibrium price and quantity as a result of these shifts?

   Part D: How might elasticity affect each of the curves and how might that affect the equilibrium? What factors contribute to elasticity in supply? Demand?

2. The Lumineers are coming to your cite and everyone wants a ticket. The numbero f tickets that fans what to buy depends no the price. When tickets are cheap, lots of people want them, but when tickets are expensive fewer people buy them. Economists estimate this demand with the equation: Qd = 500 – 5P. Meanwhile the concert organizers decide how many tickets to sell based on the price. If the price gets too low, then they won't offer many tickets, but at a higher price they're willing to sell more. The supply equation is Qs = -100 +10P.
   a. At which ticket price will the number of tickets demanded equal the number supplied?
   b. Sketch a supply-and-demand graph for this situation, labeling the equilibrium point.
   c. Suppose the Lumineers added one more day to their tour. How would this affect the equilibrium on the graph?
   d. Are the curves elastic or inelastic? Please explain.

AI-augmented Assignment Example for Intermediate Macroeconomics:

Learning Objectives

By the end of this activity, you will be able to:

- Identify the key components of a macroeconomic shock problem analyzed with the IS-MP-PC and AD-AS frameworks
- Apply economic reasoning to revise or improve practice problems
- Create and solve original problems related to short-run output and inflation dynamics
- (Optional) Analyze and critique AI-generated content for accuracy and relevance

Instructions: As a team, generate a macroeconomic shock scenario. You can use the AI of your choice (e.g., ChatGPT, Claude, Copilot, etc.) to assist you, but you are not required to do so. You want a hypothetical macroeconomic shock, and the scenario should be consistent with what we did in class, but you cannot use the same examples. Your shock should clearly map onto at least one of the parameters of the model (Hint: you may need to tell the AI the equations you are using, if AI is assisting you). You don't necessarily have to use numbers, but it is fine if you do. Create an answer key and place it inside the folder.

Intended Outcome: You want to create a scaffolded problem so that students must solve a scenario using the IS-MP-PC model first, followed by the AD-AS model. Your problem should require illustrations and explanations.

Sending a problem: Once you are confident your problem is complete and has a solution, tape the problem to the outside of the folder, put the answer inside, and pass it to another team. Teams should pass problems based on this order: 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 1

Receiving a problem: For each problem you receive, solve the problem *without* the assistance of AI and then place your solution inside the folder.

Examples of Intermediate Macroeconomics problems created by students

1. The economy is currently in long-run equilibrium with stable inflation and output at its natural level. Suddenly, households become worried about the future due to uncertainty in financial markets and high volatility. As a result, consumers cut back on spending. Firms notice declining sales and adjust their expectations.

   a) Using the IS-MP-PC model, illustrate the impact of this shock on:
   - Output (Y)
   - Inflation (π)
   - The position of IS and MP curves

   b) Using the AS-AD model, illustrate and explain the impact of this shock on:
   - AD curve
   - AS curve
   - Output
   - Price level

2. The government sends out large stimulus checks to households. As a result, consumers increase their spending on goods & services across the economy. Using the IS-MP-PC model and the AD-AS model, what happens to output and inflation in the short run if the central bank does nothing. Show and label correct shifts.

3. A sudden global pandemic spreads rapidly, leading governments to impose lockdowns and consumers to reduce spending due to health concerns. At the same time, businesses face disruptions in supply chains and labor shortages. Assume the economy is initially in short run equilibrium with output at potential and inflation at target.

   Part A. Show the IS-MP model & Explain
   Part B. Show the AD-AS Model & Explain
   Part C. What policy would you implement?

Table 1: Potential Topics to Use Send-a-Problem

Principles of Economics	Intermediate Macroeconomics
Calculating Opportunity Costs and Comparative Advantage	Measuring the Macroeconomy, including basic measures of GDP, Inflation, and Unemployment
Supply and Demand with shifting curves, and calculation of equilibrium	Solow Model
Market Structure, Elasticity	IS-MP-PC Model
Externalities and Government Intervention	AD-AS Model
AD-AS Model	